



EcoSort

AI Waste Segregation Assistant

A decision-support tool that tells anyone, instantly, which bin their waste belongs in — using AI vision and a conversational recycling assistant.

SDG 12 · Responsible Consumption

SDG 11 · Sustainable Cities

SDG 13 · Climate Action

 Live demo: codex-ts.github.io/ecosort · Repo: github.com/codex-ts/ecosort

STUDENT

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INTERNSHIP

TM1B AI for Sustainability

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1 • Problem Statement

How might we use AI to help everyday people sort their waste correctly, so that household recycling and composting can become more accurate, less contaminated, and more sustainable?

The real-world problem. Source segregation fails not because people don't care, but because they don't know where each item goes. Rules are inconsistent and counter-intuitive: a greasy pizza box is half-recyclable, a battery is hazardous, soft-plastic bags jam recycling machines, and a "recyclable" symbol doesn't mean your city accepts it. The result is **wish-cycling** — putting the wrong thing in the right bin — which contaminates entire batches and sends otherwise-recyclable material to landfill.

The scale. India generates ~62 million tonnes of municipal solid waste a year, and much that could be recycled or composted is lost to contamination at the very first step: the household bin. Contamination in mixed recycling streams commonly runs 15–25%.

2 • Who Is Affected (Target Users)

- **Households & students** who want to do the right thing but guess wrong.
- **Municipalities / urban local bodies** burdened by contaminated, unsortable waste.
- **Informal waste pickers** exposed to hazards (broken glass, batteries, e-waste).
- **The environment** — landfill methane from buried organics; wasted embodied energy & water in mis-sorted recyclables.

3 • Why AI Is Needed

- **Multimodal perception** — recognising an item from a photo, not a dropdown.
- **Instant, personalised guidance** — natural-language answers to "what about this?"
- **Scale & access** — the same expert advice for everyone, in plain language, free.
- **Explainability** — a confidence score and the reasoning behind each verdict.

4 • AI Solution Overview

EcoSort is a lightweight web app with three AI-driven parts:

Component	What it does	AI element
 Vision Classifier	Photograph a waste item → structured result: category, correct bin, prep steps, decomposition time, impact, confidence, tip.	Multimodal vision classification (Claude)
 Recycling Assistant	Answers tricky contextual questions ("greasy pizza box?", "old medicines?").	Conversational AI grounded in a waste KB
 Impact Tracker	Transparent, clearly-estimated tally of items sorted & CO ₂ avoided.	Behavioural nudge

Two modes for responsible, low-cost deployment

Mode	How it runs	Powered by
Demo (default)	Open the app — no key, no cost, offline, private.	Curated knowledge base (SWM 2016) + keyword reasoning
Live AI (optional)	Add an API key, run the backend.	Real Claude vision + chat model

A carefully engineered system prompt forces the live model to return the **same structured output** as the demo, so the experience is seamless either way — and AI is only invoked when you choose (e.g. for testing).

5 • Responsible AI Considerations

Principle	How EcoSort addresses it
Transparency	Every result shows a confidence % and a source label (knowledge base vs. live model).
Fairness	Rules drawn from public SWM 2016 standards — not assumptions about a user's location or income.
Ethics & Honesty	"Conditional" items flagged with "check locally"; no medical/hazardous advice beyond standard disposal.
Privacy	In demo mode, photos are processed in-browser and never uploaded or stored.
Scope safety	The assistant redirects off-topic questions and stays within waste/sustainability guidance.

6 • AI Elements & Tools Used

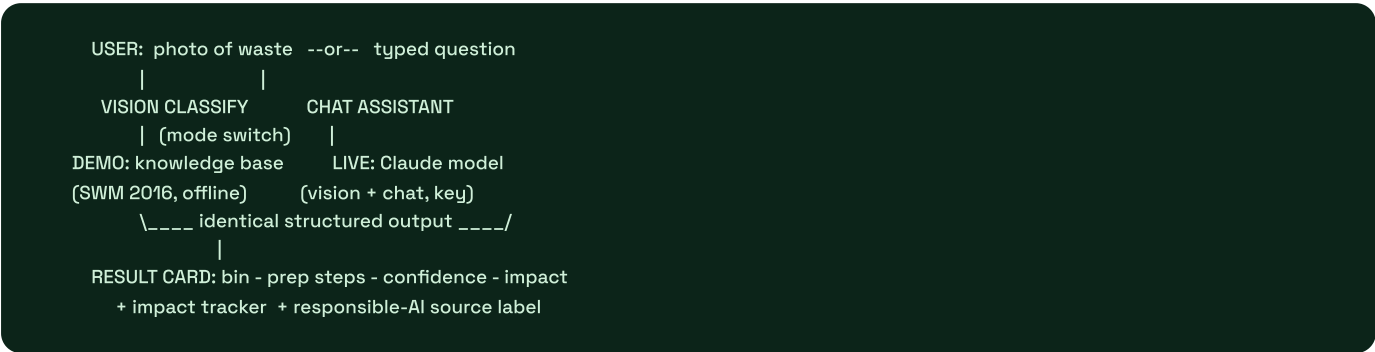
- **Multimodal vision classification** — Claude vision model identifies the item from a photo.
- **Conversational AI** — chat assistant for free-form recycling questions.
- **Prompt engineering** — structured system prompt enforcing explainable JSON output + Responsible-AI behaviour.
- **Knowledge-grounded fallback** — offline KB built on SWM 2016 for accuracy without a model call.
- **Stack** — vanilla HTML/CSS/JS front end · zero-dependency Node backend · Anthropic Messages API.

Prompt logic (the AI element)

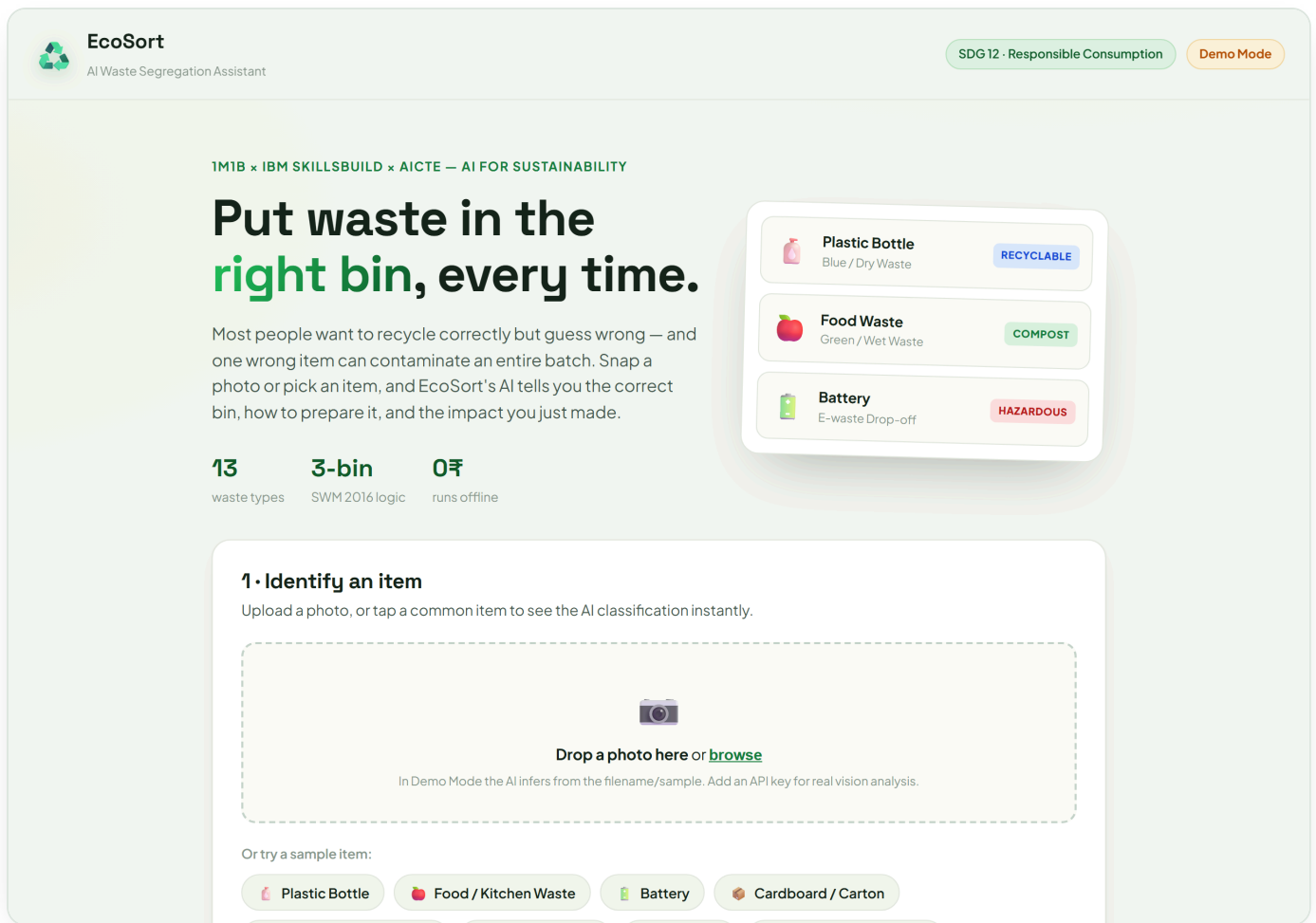
```
SYSTEM (classify): "You are EcoSort, a waste-segregation vision assistant.
Identify the single main waste item and reply with ONLY JSON:
{ name, emoji, category,
  recyclable: yes|compost|no|special|conditional|reuse,
  bin (SWM 2016 colours), binColor, confidence 0-1,
  prep[], decompose, co2, tip }.
If unclear, lower confidence and say so.
Never give medical/hazardous advice beyond standard disposal."

USER: [image] + "Classify this waste item."
```

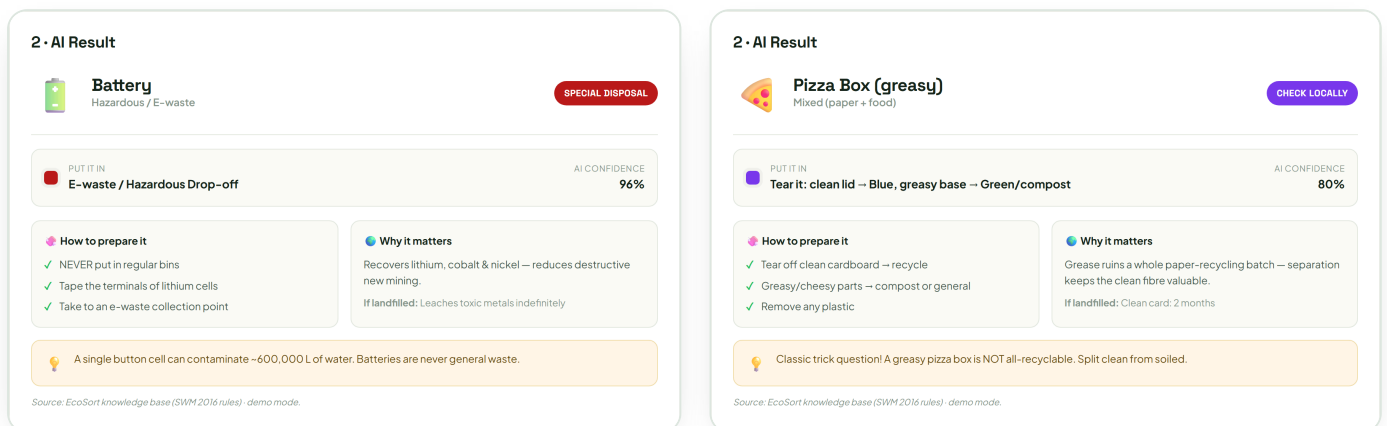
7 • Workflow Diagram



8 • Prototype / Demo



Landing page + photo/item classifier (demo mode badge visible).



Hazardous item — battery → E-waste drop-off.

"Trick" conditional item — greasy pizza box.

How do I dispose of old batteries?



Batteries are never general waste **HAZARDOUS**. A single button cell can contaminate ~600,000 L of water. Batteries are never general waste. Steps: never put in regular bins; tape the terminals of lithium cells; take to an e-waste collection point. Take them to an e-waste collection point, a battery take-back box at an electronics store, or a municipal hazardous-waste drop-off.

What can I compost at home?



You can compost most **kitchen organics** **COMPOSTABLE**: fruit & veg peels, coffee grounds, tea leaves, eggshells, and garden trimmings. **Avoid** meat, dairy, oily food and pet waste in a home compost (they smell and attract pests). Composting matters because food in landfill releases methane — ~28x more warming than CO₂.

Is a greasy pizza box recyclable?

How do I dispose of old batteries?

What can I compost at home?

Type your question...



Conversational recycling assistant answering real questions.



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Demo Mode

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Your session impact

A small, honest nudge — every correctly-sorted item counts.

2

items sorted

1

diverted from landfill

0.3 kg

est. CO₂ avoided

9 • Impact Statement

What changes if this is implemented?

- Fewer mis-sorted items → lower contamination → more material actually recycled/composted.
- Organics diverted to compost → less landfill methane (~28x more warming than CO₂ over 100 years, ~80x over 20).
- Hazardous items kept out of general bins → safer waste pickers, less soil/water pollution.

Who benefits and how?

- **Households** gain confidence and avoid rejected collections.
- **Municipalities** receive cleaner, more valuable, recyclable streams.
- **Waste pickers** face fewer hazards.
- **The planet** — every correctly-sorted item saves the embodied energy, water & carbon of new material.

Path to scale

Start as a campus/community web tool → integrate region-specific bin rules → add local languages → partner with municipal apps for point-of-disposal guidance.

EcoSort demonstrates that AI for sustainability doesn't have to be big or complex — it has to be clear, honest, and useful at the exact moment a person makes a decision. · Educational prototype; disposal rules vary by municipality — always check local guidelines.